

ADVANCED QUANTUM MECHANICS

Fundamentals of quantum mechanics



TECHNISCHE
UNIVERSITÄT
DARMSTADT

Claudiu Genes, Institute for Applied Physics; TU Darmstadt, winter semester 2025/2026

LECTURE 1: Quantum operators and states. The Schrödinger equation

We will begin by revisiting some fundamental concepts of quantum mechanics, including the Schrödinger equation in Dirac notation, its physical meaning, time evolution, and the position representation. To make these ideas concrete, we will work through simple examples involving a particle with only two possible quantum states. The plan is as follows:

- Discuss the physical regimes where quantum mechanics becomes essential
- Introduce the Dirac notation of abstract kets and bras
- Formulate the Schrödinger equation in Dirac notation and derive its time evolution
- Distinguish between time-dependent and time-independent Hamiltonians and explain the need for time ordering
- Solve explicitly the time evolution of a driven two-level quantum system
- Express the Schrödinger equation in the position representation
- Analyze the dynamics of a free particle in one spatial dimension

The realm of quantum

The *de Broglie wavelength* expresses the wave nature of any particle with mass m and momentum p . According to de Broglie's hypothesis, every moving particle is associated with a wavelength

$$\lambda = \frac{h}{p},$$

where h is Planck's constant. This relation connects two apparently different aspects of matter: the classical property of motion (momentum) and the quantum property of wave behavior.

When the momentum p is large (for example, for macroscopic objects), the wavelength λ becomes extremely small, so that wave effects such as interference or diffraction are completely negligible. In this limit, the laws of classical mechanics provide an excellent description of motion.

However, when the momentum is small, or the mass is very light, the wavelength λ becomes comparable to the size of the system under study. In that case, the wave properties of matter become important, and the motion must be described quantum mechanically.

Examples:

- **Electrons in atoms:** For an electron bound to a nucleus, the de Broglie wavelength is of the order of the atomic size ($\lambda \sim 10^{-10}$ m). The standing-wave condition for the electron's wave function leads directly to the quantization of atomic energy levels.

- **Electrons in semiconductor chips:** In modern nanostructures and transistors, electrons confined to regions of a few nanometers have de Broglie wavelengths of comparable size ($\lambda \sim 1\text{--}10\text{ nm}$). Their wave nature gives rise to phenomena such as quantum tunneling between potential barriers, discrete energy levels in quantum dots, and interference effects in nanoscale circuits.
- **Neutron diffraction:** Slow neutrons have wavelengths comparable to the spacing between crystal planes. When directed onto a crystal, they produce diffraction patterns, demonstrating their wave-like nature.
- **Cold atoms:** When atoms are cooled to very low temperatures, their momenta decrease and their de Broglie wavelengths increase. Under these conditions, many atoms can occupy the same quantum state, forming a Bose–Einstein condensate.
- **Macroscopic limit:** For everyday objects, such as a baseball of mass 0.1 kg moving at 10 m/s , the de Broglie wavelength is around 10^{-34} m , far too small to have any observable quantum effects. Classical mechanics thus becomes an excellent approximation.

Dirac notation. Operators, bra and kets. Time independent evolution

In quantum mechanics, the physical state of a system is described by a **state vector** (or *ket*) $|\psi(t)\rangle$, which lives in a complex vector space called a *Hilbert space*, denoted by $\mathcal{H}$. The ket contains all the information about the system at a given time t . You can think of it as a vector, as we will see that the matrix formulation of quantum mechanics is very appealing especially when it comes to numerical simulations.

To every ket $|\psi\rangle$ there corresponds a *bra* $\langle\psi|$ (a row vector), which is its Hermitian conjugate (or complex transpose). The inner product between two states is written as $\langle\phi|\psi\rangle$ and represents a complex number. The quantity $|\langle\phi|\psi\rangle|^2$ gives the probability of finding the system prepared in state $|\psi\rangle$ when measured in the state $|\phi\rangle$. The normalization condition $\langle\psi|\psi\rangle = 1$ expresses the fact that total probability must be one.

In Dirac notation, *operators* act on kets to produce new kets. For example,

$$\hat{A}|\psi\rangle = |\phi\rangle$$

means that the operator $\hat{A}$ transforms the state $|\psi\rangle$ into another state $|\phi\rangle$. The expectation value of an observable $\hat{A}$ in the state $|\psi\rangle$ is given by

$$\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle.$$

For the observable to correspond to a real measurable quantity, the operator must be *Hermitian*, i.e. $\hat{A}^\dagger = \hat{A}$.

The time evolution of the quantum state is governed by the *time-dependent Schrödinger equation* (TDSE),

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}(t)|\psi(t)\rangle, \quad (1)$$

where $\hat{H}(t)$ is the Hamiltonian operator, representing the total energy of the system (kinetic plus potential). Given an initial condition $|\psi(0)\rangle$ at $t = 0$, the TDSE determines the state at any later time.

The Hermitian nature of the Hamiltonian, $\hat{H}^\dagger = \hat{H}$, ensures that the norm of the state is preserved during evolution:

$$\frac{d}{dt}\langle\psi(t)|\psi(t)\rangle = 0.$$

This means that probabilities are conserved, and the time evolution is *unitary*. The norm is then fixed to unity $\langle\psi(t)|\psi(t)\rangle = 1$.

If the Hamiltonian is *time-independent*, the TDSE has a simple and elegant solution.

General solution of the time dependent Schrödinger equation in Dirac notation (for time-independent Hamiltonians)

The formal time-evolution operator is

$$\hat{U}(t) = e^{-\frac{i}{\hbar} \hat{H} t},$$

so that

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle = e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle.$$

This is straightforward, the problem is evaluating this exponential of an operator.

This operator is unitary, $\hat{U}^\dagger \hat{U} = \mathbb{1}$, ensuring norm conservation. The exponential of an operator $\hat{A}$ is defined through its power-series expansion:

$$e^{\hat{A}} = \sum_{n=0}^{\infty} \frac{1}{n!} \hat{A}^n = \mathbb{1} + \hat{A} + \frac{1}{2!} \hat{A} \hat{A} + \frac{1}{3!} \hat{A} \hat{A} \hat{A} + \dots$$

This definition is completely analogous to the Taylor expansion of the exponential function for real or complex numbers, but here the powers of $\hat{A}$ represent repeated operator multiplication.

If $\hat{H}$ has a discrete spectrum $\hat{H} |E_n\rangle = E_n |E_n\rangle$, and can be written as $\hat{H} = \sum_n E_n |E_n\rangle \langle E_n|$, then the initial state can be expanded in this eigenbasis as

$$|\psi(0)\rangle = \sum_n c_n |E_n\rangle, \quad c_n = \langle E_n | \psi(0) \rangle.$$

Substituting this into the time-evolution formula gives

$$|\psi(t)\rangle = \sum_n c_n e^{-\frac{i}{\hbar} E_n t} |E_n\rangle.$$

Each energy eigenstate evolves only by acquiring a phase factor proportional to its energy, while the probabilities $|c_n|^2$ remain constant in time. Let's now summarize below:

Solving the Schrödinger equation

Solving the Schrödinger equation essentially means finding a basis in which the Hamiltonian $\hat{H}$ is diagonal. In that basis, the eigenvalue equation $\hat{H} |E_n\rangle = E_n |E_n\rangle$ defines the stationary states (energy eigenstates) of the system.

Once this spectral decomposition

$$\hat{H} = \sum_n E_n |E_n\rangle \langle E_n|$$

is known, any initial state can be expanded as

$$|\psi(0)\rangle = \sum_n c_n |E_n\rangle, \quad c_n = \langle E_n | \psi(0) \rangle.$$

The time evolution then becomes simple:

$$|\psi(t)\rangle = \sum_n c_n e^{-\frac{i}{\hbar} E_n t} |E_n\rangle.$$

Thus, solving the Schrödinger equation amounts to:

- (i) Finding the eigenvalues E_n and eigenvectors $|E_n\rangle$ of $\hat{H}$;
- (ii) Determining the initial coefficients c_n i.e., the overlap of the initial state with the eigenvectors;
- (iii) Determining how the coefficients c_n acquire phase factors $e^{-\frac{i}{\hbar} E_n t}$ in time.

Unitary evolution operator - the time dependent case

When the Hamiltonian depends explicitly on time, $\hat{H}(t)$, the Schrödinger equation remains linear but becomes more subtle to solve, because the Hamiltonian operator at different times may not commute. This means that in general

$$[\hat{H}(t_1), \hat{H}(t_2)] \neq 0,$$

so we cannot simply pull $\hat{H}$ out of the time integral and write the solution as a single exponential, as we did in the time-independent case. The reason is that non-commuting operators do not obey the simple rule $e^{A+B} = e^A e^B$; instead, the order in which they act matters.

To account for this, one defines the *time-ordered exponential*, which ensures that operators corresponding to later times act to the left of those at earlier times. The formal solution to the Schrödinger equation then reads

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, \quad U(t) = \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_0^t \hat{H}(t') dt' \right].$$

Here the time-ordering operator $\mathcal{T}$ acts on any product of Hamiltonians as

$$\mathcal{T} [\hat{H}(t_1) \hat{H}(t_2)] = \begin{cases} \hat{H}(t_1) \hat{H}(t_2), & t_1 > t_2, \\ \hat{H}(t_2) \hat{H}(t_1), & t_2 > t_1, \end{cases}$$

and similarly for higher-order products. This prescription guarantees that the operator corresponding to the latest time always stands to the left, matching the physical idea that evolution proceeds in chronological order.

Expanding the time-ordered exponential as a series gives the *Dyson expansion*:

$$U(t) = \mathbb{1} + \left(-\frac{i}{\hbar} \right) \int_0^t dt_1 \hat{H}(t_1) + \left(-\frac{i}{\hbar} \right)^2 \int_0^t dt_1 \int_0^{t_1} dt_2 \hat{H}(t_1) \hat{H}(t_2) + \dots$$

Each term involves an ordered integration over time variables such that $0 \leq t_n \leq \dots \leq t_1 \leq t$. The restriction of the integration domain precisely enforces the correct temporal order of the operators.

The time ordering property is a simple one and you will understand it better during the exercise session where you will solve a set of classical coupled linear differential equations with time dependent coefficients and convince yourself of the necessity of time ordering.

Sanity check: the time independent case – If the Hamiltonian is actually *time independent*, $\hat{H}(t) = \hat{H}$, then all Hamiltonians at different times commute trivially, and the time-ordering operator $\mathcal{T}$ becomes unnecessary. In this case, the Dyson expansion simplifies considerably.

The first-order term reads

$$U^{(1)}(t) = \left(-\frac{i}{\hbar} \right) \int_0^t dt_1 \hat{H} = \left(-\frac{i}{\hbar} \right) t \hat{H}.$$

The second-order term becomes

$$U^{(2)}(t) = \left(-\frac{i}{\hbar} \right)^2 \int_0^t dt_1 \int_0^{t_1} dt_2 \hat{H} \hat{H} = \left(-\frac{i}{\hbar} \right)^2 \hat{H}^2 \int_0^t dt_1 \int_0^{t_1} dt_2 = \left(-\frac{i}{\hbar} \right)^2 \frac{t^2}{2!} \hat{H}^2.$$

It is easy to see that at order n , the nested integrals give

$$\int_0^t dt_1 \int_0^{t_1} dt_2 \dots \int_0^{t_{n-1}} dt_n = \frac{t^n}{n!},$$

so that the entire series reads

$$U(t) = \mathbb{1} + \sum_{n=1}^{\infty} \left(-\frac{i}{\hbar} \right)^n \frac{t^n}{n!} \hat{H}^n = \exp \left[-\frac{i}{\hbar} t \hat{H} \right].$$

Solving the time evolution of a Schrödinger equation for a two-level system

Let us assume a system which has only two states indexed by $|0\rangle$ and $|1\rangle$. This is what would be generically called a qubit. Let us assume a Hamiltonian of the following form $\hat{H} = \hbar \Omega \sigma_x$, where

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is a Pauli matrix. Its action on the basis states is

$$\sigma_x |0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle,$$

$$\sigma_x |1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle.$$

Notice that we didn't specify what the spatial form of the two level system is. This could be the ground and first excited state of an electron in an atom, two spin states of an electron in an externally applied magnetic field and so on. The reason for not specifying this is that we only care at the moment about the time evolution and how the system makes transitions between the two states. The actual spatial form of the state vectors will be tackled next when we refer to the position representation.

We will proceed based on the procedure we highlighted before:

(i) **Diagonalize the Hamiltonian.**

Since the eigenset of the Hamiltonian is the same as that of σ_x , we solve for its eigenvalues first:

$$\det(\sigma_x - \lambda I) = \det \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 - 1 = 0 \Rightarrow \lambda_{\pm} = \pm 1.$$

To find the eigenvector for $\lambda = +1$, we solve $(\sigma_x - I)\mathbf{v} = 0$:

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \Rightarrow v_1 = v_2.$$

Normalized, this gives

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Similarly one finds:

$$|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Therefore, the energy eigenvalue equations

$$\hat{H} |\pm\rangle = E_{\pm} |\pm\rangle$$

give

$$E_+ = +\hbar\Omega, \quad E_- = -\hbar\Omega,$$

and the spectral resolution is

$$\hat{H} = E_+ |+\rangle\langle+| + E_- |-\rangle\langle-|.$$

(ii) **Project the initial state and evolve each component.** Take the initial state $|\psi(0)\rangle = |0\rangle$. Its overlaps with the energy eigenvectors are

$$c_+ = \langle+|\psi(0)\rangle = \langle+|0\rangle = \frac{1}{\sqrt{2}}, \quad c_- = \langle-|\psi(0)\rangle = \langle-|0\rangle = \frac{1}{\sqrt{2}}.$$

The solution of the Schrödinger equation in the energy basis is

$$|\psi(t)\rangle = c_+ e^{-\frac{i}{\hbar} E_+ t} |+\rangle + c_- e^{-\frac{i}{\hbar} E_- t} |-\rangle = \frac{1}{\sqrt{2}} e^{-i\Omega t} |+\rangle + \frac{1}{\sqrt{2}} e^{+i\Omega t} |-\rangle.$$

Re-expressing $|\pm\rangle$ in the original basis, we get the exact time evolution of the state vector

$$|\psi(t)\rangle = \cos(\Omega t) |0\rangle - i \sin(\Omega t) |1\rangle.$$

This shows that the Hamiltonian we have considered generates flips between the zero and unity states. This is what one would call a single qubit in quantum information or quantum computing.

Introducing the position representation and the wave function

In one spatial dimension, the position representation is constructed by introducing the continuous set of kets $\{|X\rangle\}$, which satisfy

$$\hat{x} |X\rangle = X |X\rangle,$$

so that each $|X\rangle$ is an eigenstate of the position operator $\hat{x}$ with eigenvalue $X \in \mathbb{R}$. These kets form a complete orthonormal basis in the sense that

$$\langle X|X'\rangle = \delta(X - X'), \quad \int_{-\infty}^{+\infty} dX |X\rangle\langle X| = \mathbb{1}.$$

Notice that we denoted on purpose numbers by X to distinguish from the [position operator](#) $\hat{x}$.

Position representation and completeness

The completeness relation

$$\int_{-\infty}^{+\infty} dX |X\rangle\langle X| = \mathbb{1}$$

allows us to represent any state $|\psi\rangle$ as a continuous superposition of position eigenstates:

$$|\psi\rangle = \mathbb{1} |\psi\rangle = \int_{-\infty}^{+\infty} dX |X\rangle\langle X| |\psi\rangle = \int_{-\infty}^{+\infty} \psi(X) |X\rangle dX, \quad \psi(X) = \langle X|\psi\rangle.$$

The function $\psi(X)$ is called the [wavefunction](#) and contains the complete physical information about the system. The modulus squared, $|\psi(X)|^2$, represents the probability density for finding the particle at position X .

The action of the fundamental operators in this representation follows directly from their definitions. Using the completeness relation, we find

$$\langle X|\hat{x}|\psi\rangle = \int \langle X|\hat{x}|X'\rangle \langle X'|\psi\rangle dX' = \int X' \delta(X - X') \psi(X') dX' = X \psi(X),$$

so the position operator acts as multiplication by X :

$$\hat{x} \longrightarrow X.$$

For the momentum operator, we use the canonical commutation relation

$$[\hat{x}, \hat{p}] = i\hbar \mathbb{1}.$$

Starting from the commutation relation

$$[\hat{x}, \hat{p}] = i\hbar \mathbb{1},$$

we act between position eigenstates $\langle X|$ and $|\psi\rangle$:

$$\langle X| [\hat{x}, \hat{p}] |\psi\rangle = i\hbar \langle X|\psi\rangle.$$

Insert the completeness relation $\mathbb{1} = \int dX' |X'\rangle\langle X'|$ between the operators:

$$\langle X|\hat{x}\hat{p}|\psi\rangle = \int dX' \langle X|\hat{x}|X'\rangle \langle X'|\hat{p}|\psi\rangle, \quad \langle X|\hat{p}\hat{x}|\psi\rangle = \int dX' \langle X|\hat{p}|X'\rangle \langle X'|\hat{x}|\psi\rangle.$$

Since $\hat{x} |X'\rangle = X' |X'\rangle$ and $\langle X| \hat{x} = X \langle X|$, these become

$$\int dX' X \langle X| \hat{p} |X'\rangle \langle X'| \psi \rangle - \int dX' X' \langle X| \hat{p} |X'\rangle \langle X'| \psi \rangle = i\hbar \langle X| \psi \rangle.$$

Because this must hold for any $|\psi\rangle$, the integrands must be equal:

$$(X - X') \langle X| \hat{p} |X'\rangle = i\hbar \delta(X - X').$$

This is the distributional equation that determines the matrix elements of the momentum operator in the position basis with solution

$$\langle X| \hat{p} |X'\rangle = -i\hbar \frac{\partial}{\partial X} \delta(X - X').$$

Acting on an arbitrary state then gives

$$\langle X| \hat{p} |\Phi\rangle = -i\hbar \frac{\partial}{\partial X} \langle X| \Phi \rangle,$$

so that, in the position representation,

$$\hat{p} \longrightarrow -i\hbar \frac{d}{dX}.$$

The Schrödinger equation in position representation

Let us now start from the time-dependent Schrödinger equation in Dirac notation,

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle, \quad \hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}).$$

We aim at finding how to write this in the position representation, meaning that we want to find an equation for the wavefunction as a function of time and position. Project onto the position eigenbra $\langle X|$ and insert the identity $\mathbb{1} = \int_{-\infty}^{+\infty} dX' |X'\rangle \langle X'|$:

$$i\hbar \frac{\partial}{\partial t} \langle X| \psi(t) \rangle = \langle X| \hat{H} |\psi(t) \rangle = \int_{-\infty}^{+\infty} dX' \langle X| \hat{H} |X'\rangle \langle X'| \psi(t) \rangle.$$

Define the wavefunction $\psi(X, t) = \langle X| \psi(t) \rangle$. We now evaluate the Hamiltonian kernel $\langle X| \hat{H} |X'\rangle$ term by term.

Potential term. Since $V(\hat{x})$ acts multiplicatively in the position basis,

$$\langle X| V(\hat{x}) |X'\rangle = V(X) \delta(X - X').$$

Kinetic term. We evaluate

$$\langle X| \hat{p}^2 |X'\rangle = \int dY \langle X| \hat{p} |Y\rangle \langle Y| \hat{p} |X'\rangle = (-i\hbar)^2 \partial_X^2 \delta(X - X') = -\hbar^2 \partial_X^2 \delta(X - X').$$

Putting it together. Hence

$$\langle X| \hat{H} |X'\rangle = \left[-\frac{\hbar^2}{2m} \partial_X^2 + V(X) \right] \delta(X - X').$$

Insert this into the projected TDSE:

$$i\hbar \partial_t \psi(X, t) = \int_{-\infty}^{+\infty} dX' \left[-\frac{\hbar^2}{2m} \partial_X^2 + V(X) \right] \delta(X - X') \psi(X', t).$$

Using $\int dX' \delta(X - X') \psi(X', t) = \psi(X, t)$ and that ∂_X acts on X , we obtain

The Schrödinger equation in position representation

The Schrödinger equation in position representation

$$i\hbar \frac{\partial}{\partial t} \psi(X, t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial X^2} + V(X) \right] \psi(X, t).$$

This is the one-dimensional Schrödinger equation in the position representation, for a particle of mass m moving in a potential $V(X)$.

Let us now assume a separable form, which is allowed by the fact that we have a time-independent potential $V(X)$:

$$\psi(X, t) = \phi(X) T(t).$$

Substituting this into the equation gives

$$i\hbar \phi(X) \frac{dT(t)}{dt} = \left[-\frac{\hbar^2}{2m} \frac{d^2 \phi(X)}{dX^2} + V(X) \phi(X) \right] T(t).$$

Divide both sides by $\phi(X) T(t)$:

$$i\hbar \frac{1}{T(t)} \frac{dT(t)}{dt} = \frac{1}{\phi(X)} \left[-\frac{\hbar^2}{2m} \frac{d^2 \phi(X)}{dX^2} + V(X) \phi(X) \right].$$

Since the left-hand side depends only on t and the right-hand side only on X , they must both be equal to the same constant E :

$$i\hbar \frac{1}{T} \frac{dT}{dt} = E, \quad \frac{1}{\phi} \left[-\frac{\hbar^2}{2m} \frac{d^2 \phi}{dX^2} + V(X) \phi \right] = E.$$

The time equation gives

$$i\hbar \frac{dT}{dt} = E T \implies T(t) = e^{-\frac{i}{\hbar} E t}.$$

The spatial equation becomes

The stationary Schrödinger equation in position representation

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dX^2} + V(X) \right] \phi(X) = E \phi(X).$$

This is the one-dimensional stationary part of the Schrödinger equation. It requires finding the stationary wavefunctions $\phi_n(X)$ solving this equation, which also form a complete orthonormal set $\int \phi_m^*(X) \phi_n(X) dX = \delta_{mn}$.

Thus, the complete separable solution is

$$\psi(X, t) = \phi_n(X) e^{-\frac{i}{\hbar} E_n t},$$

and initial state can be expanded as

$$\psi(X, 0) = \sum_n c_n \phi_n(X), \quad c_n = \int \phi_n^*(X) \psi(X, 0) dX.$$

Each component then evolves independently in time as

$$\psi(X, t) = \sum_n c_n \phi_n(X) e^{-\frac{i}{\hbar} E_n t}.$$

This procedure—defining the position eigenkets, identifying the operator forms, and expanding in stationary eigenstates—provides the standard route for solving the Schrödinger equation in the position representation.

Example: free particle in one dimension

Note: from now on we revert to writing the position with x as we will deal with the Schrödinger equation in the position representation. We made our point that it is important to distinguish between the operator and its eigenvalues.

We will use later on, especially in the scattering theory chapter, the eigenstates of the free particle. Here we restrict to one dimension, which for a free particle, $V(x) = 0$, leads to the following stationary Schrödinger equation

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} = E \psi(x).$$

Negative energy solutions would lead to exponentially diverging wave functions at infinity; therefore, we restrict to positive energies corresponding to plane-wave solutions. Introducing the wavenumber

$$k = \pm \frac{\sqrt{2mE}}{\hbar},$$

the general solutions are plane waves (which are eigenstates of the momentum operator in the position representation, as we will see when moving to the momentum representation):

$$\phi_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx}, \quad E_k = \frac{\hbar^2 k^2}{2m}.$$

These are orthonormal in the Dirac sense (where their overlap is a delta distribution),

$$\int_{-\infty}^{\infty} \phi_{k'}^*(x) \phi_k(x) dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ik'x} e^{ikx} dx = \delta(k - k').$$

This is a system with a continuous energy spectrum: the label k indexing the eigenstates takes continuous values, both positive and negative. For a given $|k|$ the system is degenerate admitting both left and right propagating plane waves.

General solution (Fourier expansion) – An arbitrary initial state $\psi(x, 0)$ can be expanded as

$$\psi(x, 0) = \int_{-\infty}^{\infty} c(k) \phi_k(x) dk,$$

where the components are simply Fourier components

$$c(k) = \int_{-\infty}^{\infty} \phi_k^*(x) \psi(x, 0) dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} \psi(x, 0) dx.$$

Each momentum component evolves as $e^{-iE_k t/\hbar} = e^{-i(\hbar k^2/2m)t}$, yielding the full wavefunction solution of the Schrödinger equation as

$$\psi(x, t) = \int_{-\infty}^{\infty} \frac{dk}{\sqrt{2\pi}} c(k) e^{ikx} e^{-i\frac{\hbar k^2}{2m}t}.$$

References

- [1] Schwabl, F., *Quantum Mechanics*, 4th ed., Springer, Berlin / Heidelberg, 2007. <https://doi.org/10.1007/978-3-540-71933-5>
- [2] Sakurai, J. J. and Napolitano, J., *Modern Quantum Mechanics*, 2nd ed., Addison-Wesley / Pearson, 2010. <https://www.cambridge.org/core/books/modern-quantum-mechanics/AAE1925F1A0963C6124421B03D7801AE>
- [3] Cohen-Tannoudji, C., Diu, B., and Laloë, F., *Quantum Mechanics*, Volumes 1 and 2, Wiley, New York, 1977.